

Chapter 9

In Class Examples

Slides 2-3

1. If $y = x^3 - 5x^2 + 2x + 4$
 - a. Evaluate this polynomial for x values ranging from -1 to 5 in increments of 0.05. Plot the data.

```
x=-1:.05:5;  
y=x.^3-5*x.^2+2*x+4;  
plot(x,y)  
grid on
```

- b. Instead of typing out the polynomial, put the coefficients in a vector. Find the roots of the polynomial. Also extend the plot to include values of x ranging from 5 to 6 in increments of 0.05. Include this new data on the plot. Verify that the roots calculated match those found on the plot.

```
p=[1 -5 2 4];  
zeros=roots(p)  
x_values=5:.05:6;  
new_data=polyval(p,x_values)  
hold on  
plot(x_values,new_data,'m')
```

Slides 4-13

2. Download curve_data.txt from the website. This file contains time and velocity data of a projectile.
 - a. Plot the data points

```
load curve_data.txt  
x=curve_data(:,1);  
y=curve_data(:,2);  
plot(x,y,'bs')
```

- b. Use the polyfit function to curve fit the data. The data has a quadratic shape so try an order two fit. Use the coeffs from the polyfit function and evaluate the calculated polynomial. Plot the curve fitted polynomial data.

```
coeffs=polyfit(x,y,2)  
fit_data=polyval(coeffs,x)  
hold on  
plot(x,fit_data,'r')
```

Note: the polynomial used to generate this data was: $-4.905t^2 + 15.32089t + 150$. Compare these coefficients to the ones found by polyfit....they are the same!

3. Properties of the atmosphere are given below. (Download atmos_data.txt)

Altitude (m)	Temp (Celsius)	Pressure (kPa)
0	15	101.33
200	13.7	98.95
400	12.4	96.61
600	11.1	94.32
800	9.8	92.08
1000	8.5	89.88
1200	7.2	87.72
1400	5.9	85.6
1600	4.6	83.53
1800	3.3	81.49
2000	2	79.5
2200	0.7	77.55
2400	-0.59	75.63
2600	-1.89	73.76

Use interpolation to find the temperature and pressure at the following altitudes: 450 m , 1111 m, 1874 m, 2579 m.
Use both interpolation methods discussed in the lecture. Plot the tabular and interpolated data.

```
load atmos_data.txt
alt=atmos_data(:,1);
temp=atmos_data(:,2);
press=atmos_data(:,3);

coeffs_temp=polyfit(alt,temp,1)
coeffs_press=polyfit(alt,press,2)

alt_input=[450 1111 1874 2579];
temp_interp=polyval(coeffs_temp,alt_input)
press_interp=polyval(coeffs_press,alt_input)
fprintf('The temperatures are: %.2f, %.2f, %.2f, %.2f\n',temp_interp)
fprintf('The pressures are: %.2f, %.2f, %.2f, %.2f\n',press_interp)

subplot(1,2,1)
plot(alt,temp,'bs-')
hold on
plot(alt_input,temp_interp','rp-')
title('Temperature')
xlabel('Altitude (m)')
ylabel('Temp (C)')
subplot(1,2,2)
plot(alt,press,'bs-')
hold on
plot(alt_input,press_interp','rp-')
title('Pressure')
xlabel('Altitude (m)')
ylabel('Pressure (kPa)')
```

Now use interp1 to interpolate. Also determine the altitude for the following temps: 8.0, 1.0

```

load atmos_data.txt
alt=atmos_data(:,1);
temp=atmos_data(:,2);
press=atmos_data(:,3);

coeffs_temp=polyfit(alt,temp,1)
coeffs_press=polyfit(alt,press,2)

alt_input=[50 1111 1874 2759];
temp_input=[8.0 1.0];

temp_interp=interp1(alt,temp,alt_input,'linear');
press_interp=interp1(alt,press,alt_input,'spline');
alt_interp=interp1(temp,alt,temp_input,'linear');
fprintf('The temperatures are: %.2f, %.2f, %.2f, %.2f\n',temp_interp)
fprintf('The pressures are: %.2f, %.2f, %.2f, %.2f\n',press_interp)
fprintf('The altitudes are: %.2f, %.2f\n',alt_interp)
subplot(1,2,1)
plot(alt,temp,'bs-')
hold on
plot(alt_input,temp_interp,'rp-')
title('Temperature')
xlabel('Altitude (m)')
ylabel('Temp (C)')
subplot(1,2,2)
plot(alt,press,'bs-')
hold on
plot(alt_input,press_interp,'rp-')
title('Pressure')
xlabel('Altitude (m)')
ylabel('Pressure (kPa)')

```

Slides 14-16

4. Find the roots of the polynomial: $s^2 + 15s + 100$. Also find the real and imaginary parts of the roots.

```

coeffs=[1 15 100];
poles=roots(coeffs)
real(poles)
imag(poles)

```

5. Find the complex conjugate of $-3+2j$ and write an fprintf statement to display the result

```

z=-3+2j;
c=conj(z)
fprintf('The complex conjugate is %f + %fj\n',real(c),imag(c))

```

6. Test out the sym and syms commands.

```
x=sym('x');
eqn1=2*x
pause
clear all
syms x
eqn2=2*x
pause
clear all
eqn3=sym('2*x')
pause
syms y z
b=3*y+z
eqn4=2*b
```

Notice we created
the same symbolic
equation 3
different ways

7. If you are using symbolic math, MATLAB will convert decimals to symbolic fractions.

```
syms a b c x
func=a*x^2+b*x+c
3.7*func
func/(.1*x)
```

8. Turn the following polynomial into a symbolic expression. Multiply the expression by 1.5.

$$5x^4 + 10x^3 + 15x^2 + 20x + 25$$

```
p=[5 10 15 20 25];
eqn=poly2sym(p)
1.5*eqn
```

9. Convert a symbolic expression to a polynomial.

```
syms y
b=14*y^5-3*y^3+2*y
poly=sym2poly(b)
```

10. For a controls problem, a closed-loop transfer function is found to be:

$$T(s) = \left(\frac{G_1}{1 + G_1 H_1} \right) \left(G_2 + \frac{1}{G_1} \right) G_3$$

Simplify this equation.

```
syms g1 g2 g3 h1
T=(g1/(1+g1*h1))*(g2+1/g1)*g3;
simplify(T)
```

11. A closed-loop transfer function is given for a mechanical system. Simplify this expression.

$$T(s) = \frac{G_3 + G_1 G_2}{1 + [G_3 + G_1 G_2] \left[H + \frac{G_2 G_4}{G_3 + G_1 G_2} \right]}$$

```
syms g1 g2 g3 g4 h
num=g3+g1*g2;
den1=g3+g1*g2;
den2=h+(g2*g4)/(g3+g1*g2);
den=1+den1*den2;
T=num/den
den_exp=expand(den)
simplify(num/den_exp)
simplify(T)
```

These will give the same result

12. Make two substitutions in the polynomial below. 1) find the resulting polynomial when $y=2$, 2) find the value of f when $a=1$, $b=2$, $c=3$, $y=5$.

```
syms a b c y
f=a*y^2+b*y+c;
subs(f, 'y', 2)
subs(f, [a b c y], [1 2 3 5])
```

13. Using the $T(s)$ from #11, find the numerator and denominator.

```
[num den]=numden(T)
```

14. Plot the following symbolic function:

$$f(x) = e^{0.2x} \cos(x)$$

Use the default settings for the x axis and then change the size of the x axis to be $[-10, 10]$.

```
syms x
eqn=exp(.2*x)*cos(x);
subplot(1,2,1)
ezplot(eqn)
subplot(1,2,2)
ezplot(eqn, [-10, 10])
```

15. The following equations are known. Use the numden function and for loops to construct $c(t)$ equations for values of ω from 15 to 30 and values of ξ for values of 0 to 0.9.

$$T_P = \frac{\pi}{\omega\sqrt{1-\xi^2}} \quad T_S = \frac{4}{\xi\omega}$$

$$c(t) = 1 - e^{-\xi\omega t} \left(\cos(\omega\sqrt{1-\xi^2}t) + \frac{\xi}{\sqrt{1-\xi^2}} \sin(\omega\sqrt{1-\xi^2}t) \right)$$

```
syms w z t
Tp=pi/(w*sqrt(1-z^2));
Ts=4/(z*w);
[num1 den1]=numden(Tp);
[num2 den2]=numden(Ts);

for i=0:0.1:0.9
    for j=15:5:30
        den3=subs(den1,[w,z],[j,i]);
        den4=subs(den2,[w,z],[j,i]);
        c=1-exp(-1*den4*t)*(cos(den3*t)+z/(sqrt(1-z^2))*sin(den3*t))
    end
end
```

Slides 26-33

16. A vehicle's aerodynamic drag is determined in part by its cross-sectional area. Estimate the area of a car by curve fitting the data to a second order polynomial and then integrating.

width	height
0	0
4.8	18.75
9.6	24
14.4	26
19.2	24
24	18.75

```
load simpsons_car.txt
width=simpsons_car(:,1);
height=simpsons_car(:,2);
curve=polyfit(width,height,2)
sym_curve=poly2sym(curve)
area_estimate=int(sym_curve,0,24)
```

17. If P dollars is invested every year at an annual interest rate of r, compounded continuously, then after T years the amount is given by A(t).

$$A(t) = Pe^{r(T-t)}$$

Suppose you \$2000 per year in a retirement fund that pays 6% continuously compounded interest. You start saving at 25. How much will your investment be worth at age 65?

```

format longg
P=input('Enter the amount saved per year:\n');
R=input('Enter the interest rate:\n');
time=input('How many years will you contribute money:\n');
rate=R/100;
syms t
A=P*exp(rate*(time-t))
%calculate amount saved by finding the area under the curve - integrate!
saved=int(A,0,time)
money_saved=double(saved)
fprintf('You will have saved $%.2f in %d years.',money_saved,time)

```

18. The position of a train is given in the equation below. Find the velocity at $t=4$ sec. Also find the acceleration equation - does this train have constant acceleration?

$$s = \frac{1}{3}t^3 + 46t - 94.7$$

```

time=4;
s_coeffs=[1/3 0 46 -94.7];
vel_coeffs=polyder(s_coeffs);
velocity=polyval(vel_coeffs,time)
accel_coeffs=polyder(vel_coeffs)

```

19. A seismograph measures the horizontal motion of the ground during an earthquake. An engineer analyzing the data determines that for a 10s interval of time, the position is approximated by the following equation. Plot the position and velocity for this 10s interval of time.

```

syms t
s=100*cos(2*pi*t);
subplot(1,2,1)
ezplot(s,[0 10])
vel_eqn=diff(s)
subplot(1,2,2)
ezplot(vel_eqn,[0 10])

```

20. Use the **quad** numerical integration function to numerically integrate the following function.

$$-50 \sin(10x) + 3x^2 - 4x - 6$$

Also create a symbolic expression for the above function and use the **int** function to integrate. Compare the results.

```
clear all
x=0:.05:10
f=@(x)-50*sin(10*x)+3*x.^2-4*x-6
plot(x,f(x))
quad(f,0,10)
syms t
func=-50*sin(10*t)+3*t^2-4*t-6;
int_eqn=int(func,0,10)
int_value=double(int_eqn)
```

21. There is also another general numerical integration function: integral. Syntax is:

$Q = \text{integral}(\text{FUN}, A, B)$ where FUN is a function handle, A is the lower integration limit, B is the upper integration limit. Infinite integration limits can be used by using $-\text{Inf}$ or Inf . Evaluate the following integral:

$$\int_2^{\infty} \frac{x+3}{(x-1)(x^2+1)} dx$$

```
clear all
f=@(x) (x+3)./((x-1).*(x.^2+1))
int_f=integral(f,2,Inf)
```

Slides 34-37

22. Furnace temperatures were collected for two weeks. The collected data is shown below.

Week 1	952	987	947	927	951	951	948
Week 2	952	950	948	947	951	949	948

Generate a plot for each week's data. The plot title should display: variance, mean, mode, and median.

In addition a horizontal line at the mean value should be drawn.

```
temps=[952 987 947 927 951 951 948];
temps2=[952 950 948 947 951 949 948];
avg1=mean(temps)
avg2=mean(temps2)
days=[0:8];

subplot(1,2,1)
plot(temps, 'bo')
hold on
plot(days, ones(size(days))*avg1, 'r-')
stg=sprintf('Variance = %.2f\nMean = %.2f\nMode = %d\nMedian = %d', ...
    var(temps), avg1, mode(temps), median(temps))
title(stg)

subplot(1,2,2)
plot(temps2, 'rp')
hold on
plot(days, ones(size(days))*avg2, 'r-')
axis([0 8 920 990])
stg=sprintf('Variance = %.2f\nMean = %.2f\nMode = %d\nMedian = %d', ...
    var(temps2), avg2, mode(temps2), median(temps2))
title(stg)
```